Topics in Algebra Assignment 1

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Q1. We can use the Schreier Theorem. Let $H_0 = K_0 = \{0\}$, $H_1 = 20\mathbb{Z}$, $K_1 = 15\mathbb{Z}$ and $H_2 = K_2 = \mathbb{Z}$.

We have that

$$\begin{aligned} & H_0 \leq H_0 + (H_1 \cap K_1) \leq H_1 \\ \implies & \{0\} \leq \{0\} + (20\mathbb{Z} \cap 15\mathbb{Z}) \leq 20\mathbb{Z} \\ \implies & \{0\} \leq 60\mathbb{Z} \leq 20\mathbb{Z} \end{aligned}$$

and that

$$\begin{aligned} & H_1 \leq H_1 + (H_2 \cap K_1) \leq H_2 \\ \implies & 20\mathbb{Z} \leq 20\mathbb{Z} + (\mathbb{Z} \cap 15\mathbb{Z}) \leq \mathbb{Z} \\ \implies & 20\mathbb{Z} \leq 5\mathbb{Z} \leq \mathbb{Z} \end{aligned}$$

So we have the refinement

$$\{0\} \leq 60\mathbb{Z} \leq 20\mathbb{Z} \leq 5\mathbb{Z} \leq \mathbb{Z}$$

We repeat this process for K.

$$\begin{aligned} & K_0 \leq K_0 + (K_1 \cap H_1) \leq K_1 \\ \implies & \{0\} \leq \{0\} + (15\mathbb{Z} \cap 20\mathbb{Z}) \leq 15\mathbb{Z} \\ \implies & \{0\} \leq 60\mathbb{Z} \leq 15\mathbb{Z} \end{aligned}$$

and

$$\begin{aligned} & K_1 \leq K_1 + (K_2 \cap H_1) \leq K_2 \\ \implies & 15\mathbb{Z} \leq 15\mathbb{Z} + (\mathbb{Z} \cap 20\mathbb{Z}) \leq \mathbb{Z} \\ \implies & 15\mathbb{Z} \leq 5\mathbb{Z} \leq \mathbb{Z}. \end{aligned}$$

So we have the refinement

$$\{0\} \leq 60\mathbb{Z} \leq 15\mathbb{Z} \leq 5\mathbb{Z} \leq \mathbb{Z}$$

The factor groups of H are $60\mathbb{Z}/\{0\} \cong \mathbb{Z}$, $20\mathbb{Z}/60\mathbb{Z} \cong C_3$, $5\mathbb{Z}/20\mathbb{Z} \cong C_4$ and $\mathbb{Z}/5\mathbb{Z} \cong C_5$.

The factor groups of K are $60\mathbb{Z}/\{0\} \cong \mathbb{Z}$, $15\mathbb{Z}/60\mathbb{Z} \cong C_4$, $5\mathbb{Z}/15\mathbb{Z} \cong C_3$ and $\mathbb{Z}/5\mathbb{Z} \cong C_5$.

The two refinements are isomorphic.

Q2. Note that since $\mathbb{Z}_{72}$ is finite, it has a composition series.

We consider the first series

$$\{0\} \leq \langle 18 \rangle \leq \langle 3 \rangle \leq \mathbb{Z}_{72}$$

which has factor groups $\langle 18 \rangle / \{0\} \cong C_4$, $\langle 3 \rangle / \langle 18 \rangle \cong C_6$ and $\mathbb{Z}_{72} / \langle 3 \rangle \cong C_3$.

$\mathbb{Z}_{72} / \langle 3 \rangle$ is already simple and does not need further refinement.

We can insert $\langle 18 \rangle \leq \langle 9 \rangle \leq \langle 3 \rangle$, the factor groups are then $\langle 3 \rangle / \langle 9 \rangle \cong C_3$ and $\langle 9 \rangle / \langle 18 \rangle \cong C_2$.

We can also insert $\{0\} \leq \langle 36\rangle \leq \langle 18\rangle$, the factor groups are then $\langle 36\rangle / \{0\} \cong C_2$ and $\langle 18\rangle / \langle 36\rangle \cong C_2$.

We have the refinement

$$\{0\} \leq \langle 36\rangle \leq \langle 18\rangle \leq \langle 9\rangle \leq \langle 3\rangle \leq \mathbb{Z}_{72}$$

All the factor groups are simple, so this is a composition series.

Now consider the second series

$$\{0\} \leq \langle 24\rangle \leq \langle 12\rangle \leq \mathbb{Z}_{72}$$

Using the same process as used for the first series, making sure every factor group is simple, we have the refinement

$$\{0\} \leq \langle 24\rangle \leq \langle 12\rangle \leq \langle 6\rangle \leq \langle 3\rangle \leq \mathbb{Z}_{72}$$

The factor groups are $\langle 24\rangle / \{0\} \cong C_3$, $\langle 12\rangle / \langle 24\rangle \cong C_2$, $\langle 6\rangle / \langle 12\rangle \cong C_2$, $\langle 3\rangle / \langle 6\rangle \cong C_2$ and $\mathbb{Z}_{72} / \langle 3\rangle \cong C_3$.

All simple, so this is also a composition series.

Thus, by either direct comparison of the factor groups, or by the Jordan-Hölder theorem that all composition series of a group are isomorphic, the two refinements are isomorphic.

Q3. $Z(S_3)$ is the trivial subgroup, and $Z(D_4) = \{e, r^2\}$. By definition of the direct product, $Z(S_3 \times D_4) = \{z \in S_3 \times D_4 \mid \forall g \in S_3 \times D_4, zg = gz\}$. Since the group operation under direct product is component-wise,

$$Z(S_3 \times D_4) = \{(a, b) \in S_3 \times D_4 \mid \forall (c, d) \in S_3 \times D_4, (ac, bd) = (ca, db)\}$$

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This is equivalent to $Z(S_3) \times Z(D_4)$. Therefore,

$$Z(S_3 \times D_4) = \{(e_{S_3}, e_{D_4}), (e_{S_3}, r^2)\}$$

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